

Founder Population Scaling Summary

สรุปผลการทดลองขนาดประชากรเริ่มต้นในโลก sexed + inheritance ณ วันที่ 2026-05-11

This report summarizes how initial founder population size changes multi-generation persistence under the current world rules.

เอกสารนี้สรุปว่าจำนวน founder เริ่มต้นส่งผลต่อการพาสายตระกูลไปข้ามรุ่นอย่างไร ภายใต้กติกาโลกชุดปัจจุบัน

Key Findings

- The current sweet spot is 250 founders. It has mean max generation 6.6 and reaches generation 7 or more in 6/10 runs.
- 300 founders is close behind. It produces more children overall, but its mean max generation (5.8) is still lower than 250.
- 200 founders clearly improves over 50-100, but remains less robust than 250-300.
- Tiny populations fail almost immediately. Population 2 never exceeded generation 1 in the test set, and population 4 never exceeded generation 1 either.
- Increasing founders helps, but it does not fix the root problem by itself. The world is still unstable and can still collapse even in high-founder runs.

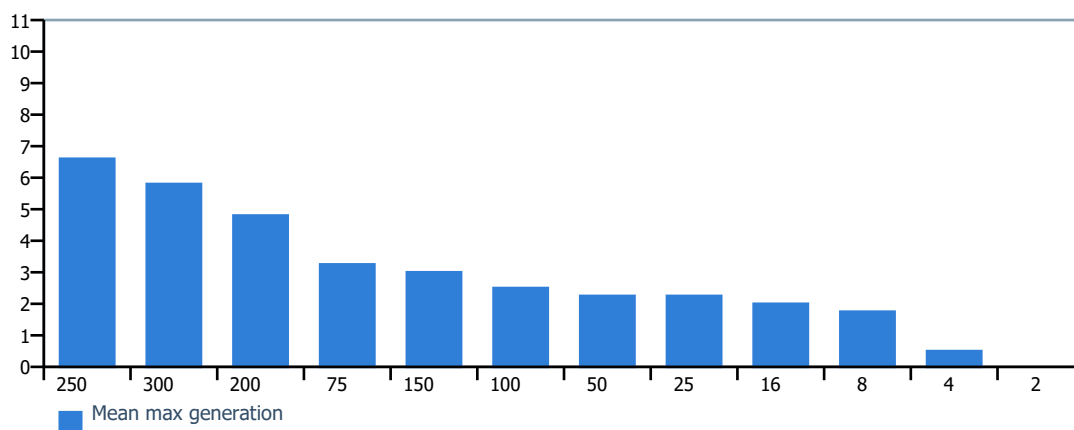
Top three populations in this stage: 250, 300, and 200 founders. Their best observed runs reached generations 10, 10, and 9 respectively.

Population Ranking

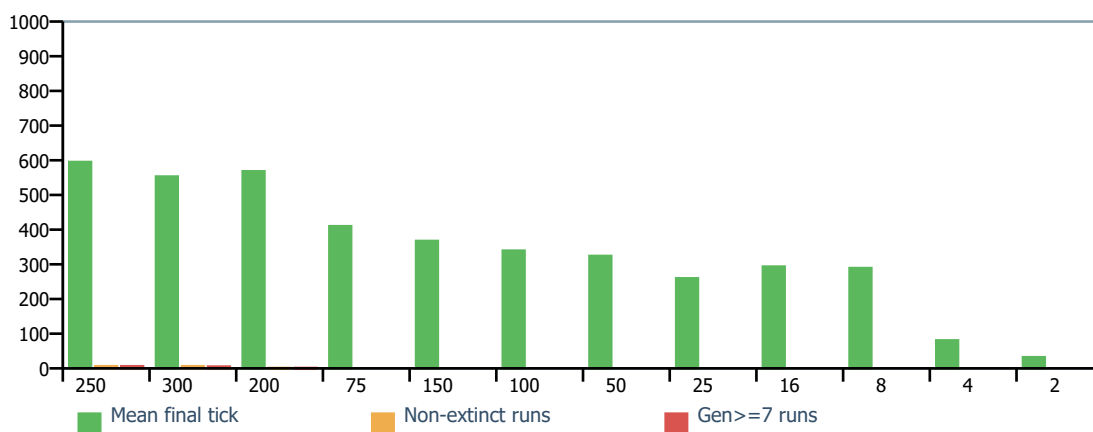
Rank	Founders	Runs	Mean max gen	Mean final tick	Gen>=5	Not extinct
1	250	10	6.6	594.9	7/10	6/10
2	300	10	5.8	552.9	6/10	6/10
3	200	10	4.8	568.1	5/10	1/10
4	75	4	3.25	409.75	1/4	0/4
5	150	4	3.0	367.25	1/4	0/4
6	100	4	2.5	339.25	0/4	0/4
7	50	4	2.25	324.0	0/4	0/4
8	25	4	2.25	259.5	0/4	0/4
9	16	4	2.0	293.25	0/4	0/4
10	8	4	1.75	289.0	0/4	0/4
11	4	4	0.5	80.5	0/4	0/4
12	2	4	0.0	32.0	0/4	0/4

Charts

Mean Maximum Generation by Founder Population



Persistence Signals by Founder Population



Note: the second chart mixes scales. Mean final tick uses the same axis as run counts, so treat it as a directional comparison rather than a precise visual magnitude comparison.

Representative Cases

Case	Mean max gen	Mean tick	Best run	Interpretation
Population 2	0.167	67.5	seed 12 -> gen 1	Cannot establish a durable reproduction chain.
Population 50	2.667	343.0	seed 12 -> gen 4	Can reach gen2-gen4, but still thins out quickly.
Population 200	4.8	568.1	seed 14 -> gen 9	Crosses the bottleneck more often, but is still unstable.
Population 250	6.6	594.9	seed 11 -> gen 10	Current best balance between depth and repeatability.
Population 300	5.8	552.9	seed 13 -> gen 10	High-output regime, but noisier than 250.

Interpretation

The current world shows a clear founder-mass threshold. Tiny populations collapse before a lineage can stabilize, mid-sized populations can establish short chains, and high populations can sometimes sustain long thin chains.

The best observed operating region is around 250 founders. That region produces the strongest combination of deep generations, longer runtime, and repeatable non-extinction within the test horizon.

The system is still not solved. Even the best region does not guarantee long-run stability, and some high-founder runs still collapse early.

This means founder count is a strong amplifier, not the full fix. The deeper root cause remains in post-generation-1 replacement dynamics, especially whether the system can keep enough adults, enough mating continuity, and enough energy to keep replacement above one.

Source Data

- C:\artificial-evolution\data\founder_population_scaling_micro_body14.json
- C:\artificial-evolution\data\founder_population_scaling_body14.json
- C:\artificial-evolution\data\founder_population_optimal_search.json